

From Topology to Combinatorics Via Computability Theory

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Ginsburg–Sands theorem

Theorem (Ginsburg & Sands, 1979).

Every infinite topological space has an infinite subspace homeomorphic to one of the following topologies on ω :

- ▶ indiscrete (only $\emptyset$ and ω are open);
- ▶ initial segment (open sets are $\emptyset$, ω , and $[0, n]$ for all $n \in \omega$);
- ▶ final segment (open sets $\emptyset$, ω , and $[n, \infty)$ for all $n \in \omega$);
- ▶ discrete (all subsets of ω are open);
- ▶ cofinite (open sets are $\emptyset$ and all cofinite subsets of ω).

We are interested in the **logical strength** of this theorem.

A foundational motivation

Reverse mathematics is motivated by a foundational question:

Question. Which axioms do we *really* need to prove a given theorem?

The question leads to the idea of the *strength* of a theorem. Which theorems does it imply? Which imply it? Which is it equivalent to?

Example. Over ZF, the axiom of choice is equivalent to Zorn's lemma.

Set theory is too strong to calibrate the strength of "ordinary theorems".

We would like results of the form

Over the theory T , theorem P implies/is equivalent to theorem Q ,
where T is *weak* enough to not prove everything, yet *robust* enough to accommodate a decent amount of basic coding and representation.

Subsystems of second-order arithmetic

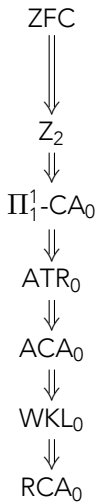
Second-order arithmetic, Z_2 , is a two-sorted theory with variables/quantifiers for **numbers** and **sets of numbers** (plus $\in$ and the usual symbols of arithmetic).

The axioms of Z_2 are those of Peano arithmetic and the **comprehension scheme**: if φ is a formula (possibly with parameters) then $\{x \in \mathbb{N} : \varphi(x)\}$ exists.

We restrict the amount of comprehension and induction to obtain **subsystems**:

- ▶ **RCA_0** : Δ_1^0 comprehension (computable sets) + Σ_1^0 induction.
- ▶ **WKL_0** : RCA_0 + every infinite binary tree $T \subseteq 2^{<\mathbb{N}}$ has an infinite path.
- ▶ **ACA_0** : RCA_0 + arithmetical comprehension.
- ▶ **ATR_0** : RCA_0 + arithmetical formulas iterated along ctbl well orders.
- ▶ **$\Pi_1^1\text{-}CA_0$** : RCA_0 + Π_1^1 comprehension.

Subsystems of second-order arithmetic



Classical reverse mathematics

Most (countable) classical mathematics can be developed within Z_2 .

Provable in RCA_0 : Baire category theorem, intermediate value theorem, Urysohn's lemma, Tietze extension theorem, Every field has algebraic closure.

Equivalent to WKL_0 over RCA_0 : Heine-Borel theorem for $[0, 1]$, Brouwer fixed-point theorem, Every commutative ring has a prime ideal.

Equivalent to ACA_0 over RCA_0 : Bolzano-Weierstrass theorem, Every vector space over $\mathbb{Q}$ has a basis, Every commutative ring has a maximal ideal.

Equivalent to ATR_0 : Comparability of well-orderings, Open/clopen determinacy for ω^ω .

Equivalent to Π^1_1 - CA_0 : Cantor/Bendixson theorem, Some higher determinacy.

Connections with computability theory

A set A is **computable from** B , written $A \leq_T B$, if $A = \Phi(B)$ for some effective procedure Φ ; i.e., knowing B , we can computably figure out A .

The **halting set** relative to a set A is

$A' = \{i \in \mathbb{N} : \text{the } i\text{th program with oracle } A \text{ halts}\}$. For all A , $A <_T A'$.

A model $\mathcal{M}$ of Z_2 is a pair $(N, \mathcal{S})$, where N is a (possibly nonstandard) model of arithmetic and $\mathcal{S} \subseteq \mathcal{P}(N)$. When $N = \omega$, $\mathcal{M}$ is called an **ω -model**.

RCA_0	ω -models closed under $\leq_T$ and $\oplus$ (disjoint union).
ACA_0	ω -models closed under the jump operator, $A \mapsto A'$.

We regard $\forall X[\phi(X) \rightarrow \exists Y\psi(X, Y)]$ theorems as **instance-solution problems**: all $X \subseteq \mathbb{N}$ s.t. $\phi(X)$ are **instances**; all $Y \subseteq \omega$ s.t. $\psi(X, Y)$ are **solutions to X** .

The strength of a theorem roughly corresponds to **how far apart in complexity** instances and solutions of a theorem are (measured in various ways).

Ramsey's theorem

Let $[X]^n$ denote all $\langle x_0, \dots, x_{n-1} \rangle \in X^n$ with $x_0 < \dots < x_{n-1}$.

RT_k^n . Every coloring $c : [\omega]^2 \rightarrow 2$ has an infinite homogeneous set.

Theorem (Specker). For $k \geq 2$, RCA_0 proves RT_k^n if and only if $n = 1$.

Theorem (Hirst). $(\forall k) RT_k^1 \leftrightarrow B\Sigma_2^0$ over RCA_0 . (Not provable in $I\Sigma_1^0$.)

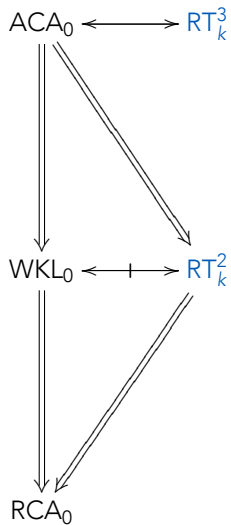
Theorem (Jockusch). For $n \geq 3$, $RT_2^n \leftrightarrow ACA_0$ over RCA_0 and $WKL_0 \not\vdash RT_2^2$.

(There exists a computable coloring $c : [\mathbb{N}]^3 \rightarrow 2$ such that every infinite homogeneous set H for c computes $\emptyset'$.)

Theorem (Seetapun). $RT_2^2 \not\vdash ACA_0$ over RCA_0 .

Theorem (Liu). $RT_2^2 \not\vdash WKL_0$ over RCA_0 . (Forcing + clever combinatorics.)

Ramsey's theorem



Stable colorings

Definition (Cholak, Jockusch, and Slaman).

- ▶ A coloring $c : [\mathbb{N}]^2 \rightarrow 2$ is **stable** if $(\forall x)[\lim_y c(x, y) \text{ exists}]$.
- ▶ **SRT₂²**: Restriction of RT_2^2 to stable colorings.

Stable colorings **non-effectively** reduce the complexity of RT_2^2 to that of RT_2^1 .

Definition. **Δ_2^0 -Subset** (a.k.a. D_2^2) is the scheme consisting of

$$(\forall x)[\varphi(x) \leftrightarrow \neg\psi(x)] \rightarrow (\exists Y \text{ infinite})[(\forall x \in Y)\varphi(x) \vee (\forall x \in Y)\neg\varphi(x)],$$

where φ and ψ are Σ_2^0 formulas.

(RT_2^1 for Δ_2^0 instances.)

Theorem (Cholak, Jockusch, & Slaman; Chong, Lempp, & Yang).

Over RCA_0 , SRT_2^2 is equivalent to Δ_2^0 -Subset.

Cohesiveness

Definition. A set X is **cohesive** for a family $\langle d_n : n \in \mathbb{N} \rangle$ of colorings $\mathbb{N} \rightarrow 2$ if

$$(\forall n)(\exists F \text{ finite})[X \setminus F \text{ is homogeneous for } d_n].$$

COH: Every family of colorings of singletons has an infinite cohesive set.

Intuitively, COH and SRT_2^2 correspond, respectively, to the **pre-process** step and **post-process** step commonly seen in most standard proofs of RT_2^2 .

- ▶ Given $c : [\mathbb{N}]^2 \rightarrow 2$, define $d_n : \mathbb{N} \rightarrow 2$ by $d(y) = c(n, y)$.
- ▶ If X is an infinite cohesive set for $\langle d_n : n \in \mathbb{N} \rangle$ then $c \upharpoonright [X]^2$ is stable.

The CJS decomposition

Theorem (Cholak, Jockusch, & Slaman).

Over RCA_0 , $\text{RT}_2^2 \leftrightarrow \text{SRT}_2^2 + \text{COH}$.

A more modern proof of Seetapun's theorem (that RT_2^2 does not imply ACA_0) takes advantage of this decomposition.

Theorem. COH admits **cone avoidance**.

(Every computable $\langle d_n : n \in \mathbb{N} \rangle$ has an infinite cohesive set $X \not\leq_T \emptyset'$.)

Theorem (Dzhafarov & Jockusch). RT_2^1 admits **strong cone avoidance**.

(Every coloring $d : \mathbb{N} \rightarrow 2$ has an infinite homogeneous set $X \not\leq_T \emptyset'$.)

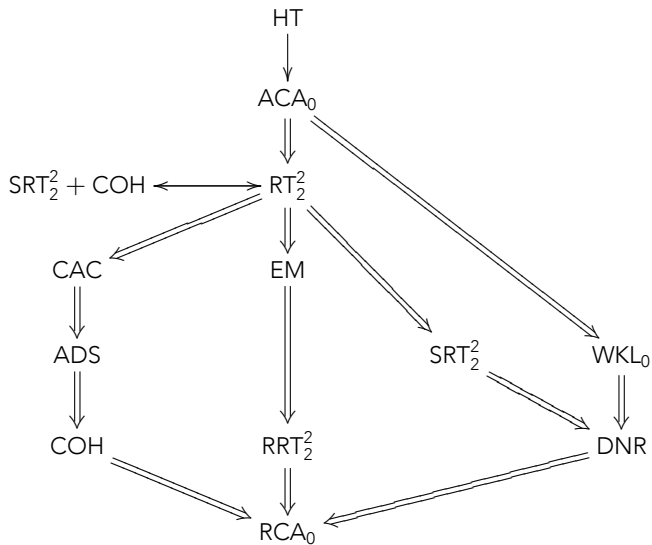
After CJS, decomposing principles into a “stable” and “cohesive” part has proved remarkably fruitful in the reverse mathematical analysis.

Other combinatorial principles

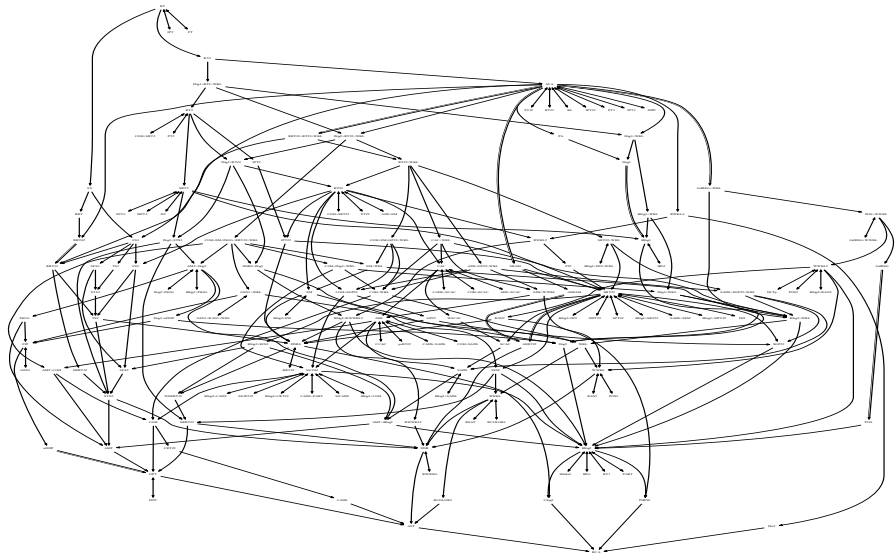
- ▶ Variations on Ramsey's theorem (polarized, rainbow, etc.).
- ▶ Thin set/free set theorems.
- ▶ Dilworth's theorem and related results about partial and linear orders.
- ▶ Erdős-Moser theorem (tournaments).
- ▶ Tree theorems (more on this later).
- ▶ Combinatorial principles related to notions of algorithmic randomness.
- ▶ Hindman's theorem.
- ▶ Carlson-Simpson lemma, dual Ramsey's theorem.

(Astor, Cholak, Chubb, Chong, Csimá, Day, Dorais, Downey, Dzhamov, Flood, Friedman, Hirschfeldt, Hirst, Jockusch, Lempp, Lerman, Marcone, McNicholl, Mileti, Patey, Shafer Shore, Simpson, Slaman, Solomon, Suggs, Towsner, Wang, Yang, ...)

The RM Zoo



The RM Zoo



Beyond combinatorics: two examples

The **atomic model theorem** (Hirschfeldt, Shore, and Slaman).

AMT/OPT: Principles formalizing that every atomic theory has atomic model.

The **finite intersection principle** (Dzhafarov and Mummert).

FIP: Every family of sets has a maximal subfamily with f.i.p.

(Classically, AMT and OPT are provable in ZF while FIP is equivalent to AC.)

Theorem.

- ▶ (Hirschfeldt, Shore, & Slaman). AMT follows from RT_2^2 and is very weak.
- ▶ (Dzhafarov & Mummert; Day, Dzhafarov, & Miller). $AMT \rightarrow FIP \rightarrow OPT$.
- ▶ (Cholak, Downey, & Igusa). FIP is equivalent to the existence of Cohen 1-generics (generics for families of Σ_1^0 -definable dense sets).

Ginsburg–Sands theorem

Theorem (Ginsburg & Sands, 1979).

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These subspaces are **minimal**, in the same sense as homogeneous sets.

Ginsburg & Sands present the theorem as an “application” of RT_2^2 .

Outline of classical proof of GS

Let X be an infinite topological space. WLOG, assume X is countable.

Define $\sim$ on X by $x \sim y$ if $x \in \text{cl}(y)$ and $y \in \text{cl}(x)$.

- ▶ If $[x]_{\sim}$ is infinite for some x , this is an infinite indiscrete subspace.

So we can assume X is T_0 . Define a partial order $\trianglelefteq$ on X by $x \trianglelefteq y$ if $x \in \text{cl}(y)$.

- ▶ There is an infinite $\trianglelefteq$ -chain/antichain. (CAC)
- ▶ Every infinite chain has an infinite ascending/descending sequence. (ADS)
- ▶ These are homeomorphic to final/initial segment topologies, respectively.

So can assume X is T_1 . If X has no infinite subspace homeomorphic to the cofinal topology, it is not difficult to construct an infinite discrete subspace.

(The construction is inductive but not computable.)

Variants of GS

Definitions. The following are defined in RCA_0 .

GS: Every infinite CSC space has an infinite subspace that is indiscrete, has the initial segment topology, has the final segment topology, is discrete, or has the cofinal topology.

wGS: Every infinite CSC space has an infinite subspace that is indiscrete, has the initial segment topology, has the final segment topology, or is T_1 .

GS $_1$: GS restricted to T_1 CSC spaces.

GS^{cl}: GS restricted to CSC spaces for which the closure operator exists.

wGS^{cl}: wGS restricted to CSC spaces for which the closure operator exists.

Relationship to the proof of Ginsburg–Sands

Principle	Measures the strength of
GS	Full Ginsburg–Sands theorem (for c.s.c. spaces)
GS^{cl}	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$
wGS	Full theorem, minus the complexity of handling the T_1 case
wGS^{cl}	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$, and of handling T_1 case
GST_1	The construction handling the T_1 case

The strength of GS

Theorem (Benham, DeLapo, Dzhafarov, Solomon, and Villano).

The following are equivalent over RCA_0 :

- ▶ ACA_0 .
- ▶ GS.
- ▶ wGS.
- ▶ For every space, the closure operator exists.

Theorem (BDDSV).

- ▶ $\text{wGS} \leftrightarrow \text{CAC}$ over RCA_0 .
- ▶ $\text{GS}^{\text{cl}} \leftrightarrow \text{GST}_1$ over RCA_0 .

Stable spaces

Definition. A T_1 CSC space $(X, \mathcal{U}, k)$ is **stable** if

$$(\forall x)[\{x\} \in \mathcal{U} \vee (\forall U \in \mathcal{U})[x \in U \rightarrow U =^* X]].$$

► SGST_1 is the restriction of GST_1 to stable spaces.

Theorem (BDDSV). Over RCA_0 , $\text{GST}_1 \leftrightarrow \text{SGST}_1 + \text{COH}$.

Definition. $\Sigma_n^0\text{-Subset}$ is the scheme consisting of

$$(\exists Y \text{ infinite})[(\forall x \in Y)\varphi(x) \vee (\forall x \in Y)\neg\varphi(x)],$$

where φ is Σ_n^0 .

Theorem (BDDSV). Over RCA_0 , $\text{SGST}_1 \leftrightarrow \Sigma_2^0\text{-Subset}$.

Non-implications

Do any of $\text{ACA}_0 \rightarrow \text{GST}_1$ and $\text{GST}_1 \rightarrow \text{RT}_2^2$ reverse? No.

Theorem (BDDSV). There is an ω -model of GST_1 that does not contain $\emptyset'$.

Corollary. GST_1 does not imply ACA_0 over RCA_0 .

This uses strong cone avoidance of RT_2^1 and the equivalence $\text{SGST}_1 \leftrightarrow \Sigma_2^0\text{-Subset}$.

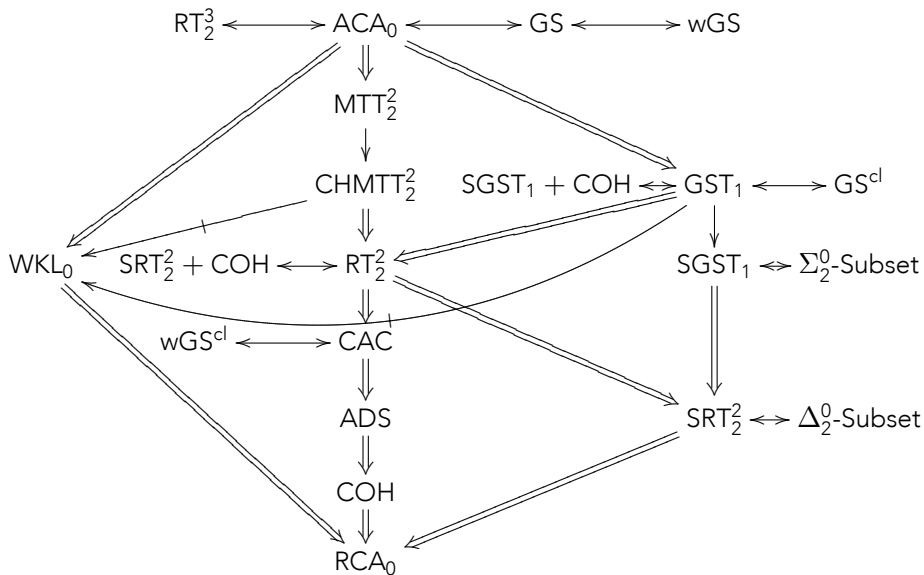
Theorem (BDDSV).

There is an ω -model satisfying $\Sigma_2^0\text{-Subset}$ but not $\Delta_2^0\text{-Subset}$.

Corollary. RT_2^2 does not imply GST_1 over RCA_0 (even ω -models).

This is a direct computability-theoretic forcing construction.

GS in the zoo



Open questions

Question. Does RCA_0 prove that every infinite effectively Hausdorff CSC space has an infinite effectively discrete subspace?

Question.

- ▶ Does Σ_2^0 -Subset imply COH?
- ▶ Does Σ_n^0 -Subset imply COH for any $n \geq 2$?

The question of whether Δ_2^0 -Subset $\rightarrow$ COH was a longstanding question in reverse math [answered by Chong, Slaman, & Yang, and by Monin & Patey].

Question. How does GST_1 relate to other principles that lie strictly in-between ACA_0 and RT_2^2 (e.g., Milliken's tree theorem)?

Question. What other animals live in the Zoo?

Thank you for your attention!
